Topological Methods in Condensed-Matter Physics

Exercise sheet 01, fall semester 2026/2027

Electronic hand-in over moodle until 04.10.2026 23:59



21.09.2026

Gergő Pintér, Levente Rózsa

A maximum of **3 complete exercises** may be submitted from this exercise sheet. The submitted exercises will be discussed in the exam.

Note: Submit the solutions in a single file.

Exercise 1: Hopf fibration

Define the map $f : \mathbb{C}^2 \supset S^3 \to \mathbb{R}^3$

$$f(\Psi) = \begin{pmatrix} \mathrm{Re}(2\bar{z}w) \\ \mathrm{Im}(2\bar{z}w) \\ |z|^2 - |w|^2 \end{pmatrix}, \tag{1}$$

where

$$\Psi = \begin{pmatrix} z \\ w \end{pmatrix} \in S^3 \subset \mathbb{C}^2, \text{ i.e. } |z|^2 + |w|^2 = 1.$$

a) Show that $\|f(\Psi)\| = 1$, that is, $f(\Psi) \in S^2 \subset \mathbb{R}^3$.

b) Show that $f(\Psi)$ can be expressed in terms of the quotient w/z, if $z \neq 0$.

c) In polar coordinates let $z = r \cdot e^{i\alpha}$, $w = s \cdot e^{i\beta}$; $r^2 + s^2 = 1$. Find $0 \leq \vartheta \leq \pi$ such that $r = \cos(\vartheta/2)$ and $s = \sin(\vartheta/2)$. Let $\varphi := \beta - \alpha$ (the relative phase). Show that

$$f(\Psi) = \begin{pmatrix} \sin\vartheta \cdot \cos\varphi \\ \sin\vartheta \cdot \sin\varphi \\ \cos\vartheta \end{pmatrix}.$$

d) Verify that $f : S^3 \to S^2$ is surjective onto S^2, and the preimage $f^{-1}(p)$ of every point $p \in S^2$ is a circle S^1 in S^3.

Exercise 2: Visualization of the Hopf circles

a) Express the stereographic projection $S^3 \setminus \{\text{point}\} \to \mathbb{R}^3$ by an explicit formula.

b) Find a 'nice' parametrization of the tori

$$T_r = \{(z, w) \in S^3 \subset \mathbb{C}^2 \mid |z| = r,\ |w| = \sqrt{1 - r^2}\}$$

c) Find a 'nice' parametrization of the Hopf circles, that is, the preimages $f^{-1}(p) \subset S^3$ of the points $p \in S^2$.

d) Verify that each Hopf circle is contained in one torus T_r, for a suitable r. What is this r?

e) By composing the parametrizations with the stereographic projection, find parametrizations for the images of the tori and Hopf circles in $\mathbb{R}^3$.

f) (Optional) Use Geogebra, Mathematica or other software to visualize the above objects in $\mathbb{R}^3$. Depict the three pictures of the state Ψ together with variable parameters, according to the description in the lecture notes.

Exercise 3: Hermitian matrices and Hopf fibration

Recall that a complex square matrix H is hermitian (or self-adjoint) if $H^\dagger = H$, where $H^\dagger = \overline{H^T}$ denotes the conjugate transpose of H. Every 2×2 hermitian matrix H can be uniquely written as a linear combination

$$H = a\sigma_0 + x\sigma_x + y\sigma_y + z\sigma_z$$

with real coefficients $a, x, y, z \in \mathbb{R}$ of the Pauli matrices

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

a) Find the eigenvalues of $H = a\sigma_0 + x\sigma_x + y\sigma_y + z\sigma_z$.

b) Find an eigenvector Ψ of unit length ($\|\Psi\| = 1$) for both eigenvalue of H.

c) Consider an H with $a = 0$ and $x^2 + y^2 + z^2 = 1$. Apply the Hopf map f (see Equation (1)) to the (normalized) eigenvectors of H. What is the result?

d) (Optional) The above matrix H is the operator of the spin measured in the (x, y, z) direction. What can we learn from the above calculation for this quantity?

e) Recall that the *expectation value* of an operator (now, a 2×2 hermitian matrix) H on the state Ψ is defined as $\Psi^\dagger \cdot H \cdot \Psi$. Show that the Hopf map f (see Equation (1)) can be expressed as

$$f(\Psi) = \begin{pmatrix} \Psi^\dagger \cdot \sigma_x \cdot \Psi \\ \Psi^\dagger \cdot \sigma_y \cdot \Psi \\ \Psi^\dagger \cdot \sigma_z \cdot \Psi \end{pmatrix}.$$

Exercise 4: Metric, topology

a) Show that every metric induces a topology. More precisely, show that the open sets of a metric space (see definition in the lecture notes) satisfy the conditions 1-3. of a topology τ (see the lecture notes). Conclude that every metric space is a topological space. Did you use all the three conditions of the definition of a metric?

b) Show that in a metric space every open ball is an open set.

c) Show that for any subset $Y \subset X$ of a topological space X, the subspace topology is really a topology, that is, it satisfies the conditions 1-3.

Exercise 5: Continuity

a) Show that the definition of the continuity for topological spaces generalizes the definition of the continuity for metric spaces (see definitions in the lecture notes). More precisely, let X and Y be metric spaces, and let $f : X \to Y$ be a map. Show that f is continuous as a map between metric spaces if and only if it is continuous as a map between the corresponding topological spaces.

b) Formally, a sequence of real numbers is a function $f : \mathbb{N} \to \mathbb{R}$. What is the condition for a sequence, as a function, to be continuous?

c) What are the continuous functions $f : X \to Y$ if X or Y is a discrete or anti-discrete topological space? Analyze the 4 cases, although, you may not give a full answer for all of them.

Exercise 6: Open and closed subsets

For the following topological spaces (X, τ) and their subsets $A \subset X$, decide whether A is an open or closed subset of X, and determine $\mathrm{cl}(A)$, $\mathrm{int}(A)$ and ∂A. (Usual metric topology means the Euclidean topology, so the default one on $\mathbb{R}^n$.)

a) $X = \mathbb{R}$, τ is its usual metric topology, $A = [0; 1)$ is an interval.

b) $X = [0; 5]$ is an interval, τ is its usual metric topology, $A = [0; 1)$ is an interval.

c) $X = \mathbb{R}$, τ is the discrete topology, $A = [0; 1)$.
d) $X = \mathbb{R}$, τ is the anti-discrete topology, $A = [0; 1)$.
e) $X = A = [0; 1)$ with the metric topology.
f) $X = \mathbb{R}$ with the usual metric topology, $A = (0; 1)$.
g) $X = \mathbb{R}^2$ with the usual metric topology, $A = (0; 1) \times \{0\}$.
h) $X = \mathbb{R}$ with the usual metric topology, $A = \mathbb{Q}$ is the set of rational numbers.
i) $X = \mathbb{R}$ with the usual metric topology, $A = \{(1 + \frac{1}{n})^n \mid n \in \mathbb{N}^+\}$.
j) $X = [-1; 3) \cup [5; 6]$ with the usual metric topology, $A = [5; 6]$.
k) Construct your own tricky example!

Exercise 7: Homeomorphic spaces

Show that the topological spaces X and Y are homeomorphic. Find an explicit homoemorphism. (X and Y are subspaces of $\mathbb{R}$ or the plane $\mathbb{R}^2$, endowed with metric topology.)

a) $X = [a, b]$, $Y = [c, d]$ (where $a < b$ and $c < d$ are real numbers).
b) $X = (a, b)$ (where $a < b$), $Y = \mathbb{R}$.
c) $X = S^1 \setminus \{\text{point}\}$ (a circle minus one point), $Y = \mathbb{R}$.
d) $X = S^1$, $Y = \partial([-1; 1] \times [-1; 1])$ is the boundary line of a square.
e) $X = D^2$ is a closed disc, $Y = [-1; 1] \times [-1; 1]$ is a square.
f) $X = \breve{B}^n$ (open ball), $Y = \mathbb{R}^n$.
g) $X = \breve{B}^n$, $Y = S^n \setminus \{\text{point}\}$.
h) $X = S^1 \times (0; 1)$, $Y = \mathbb{R}^2 \setminus \{0\}$.
i) Construct your own example!

Exercise 8: Graph of a map

Show that the graph of a continuous map is homeomorphic to the domain of the map! In more detail:
Recall that the *graph* of a map $f : X \to Y$ is defined as

$$\text{graph}(f) = \{(x, f(x)) \in X \times Y\}.$$

If X and Y are topological spaces, then $\text{graph}(f)$ inherits a subspace topology from the product topological space $X \times Y$.

a) Show that the projection $g : \text{graph}(f) \to X$, $g(x, f(x)) = x$ is always a continuous map, independently of the continuity of f.
b) Show that if f is continuous, then $\text{graph}(f)$ is homeomorphic to X!

Exercise 9: Evanescent electromagnetic waves

Consider the interference pattern of two electromagnetic plane waves of the same frequency ω,

$$\boldsymbol{E} = \text{Re}\left(\boldsymbol{E}_1 \mathrm{e}^{\mathrm{i}(\omega t - \boldsymbol{k}_1 \boldsymbol{r})} + \boldsymbol{E}_2 \mathrm{e}^{\mathrm{i}(\omega t - \boldsymbol{k}_2 \boldsymbol{r})}\right),$$

and similarly for the magnetic field $\boldsymbol{B}$.

a) Assume that the waves travel along the z direction with the same amplitude $E_1^z = E_2^z = E_0/2 \in \mathbb{R}$ and wave vector $k_1^z = k_2^z = k^z$. In the xy plane they travel oppositely to each other $k_1^x = -k_2^x = k^x, k_1^y = -k_2^y = k^y$, and create an interference pattern. Introduce the notations $k_0^2 = k_\parallel^2 + (k^z)^2 =$

$(k^x)^2 + (k^y)^2 + (k^z)^2$ and $\omega^2 = k_0^2 c^2$. Consider Maxwell's equations in vacuum,

$$\begin{aligned}\boldsymbol{\nabla} \cdot \boldsymbol{E} =& 0,\\ \boldsymbol{\nabla} \times \boldsymbol{E} =& -\partial_t \boldsymbol{B},\\ \boldsymbol{\nabla} \cdot \boldsymbol{B} =& 0,\\ \boldsymbol{\nabla} \times \boldsymbol{B} =& \frac{1}{c^2}\partial_t \boldsymbol{E}.\end{aligned}$$

Find a transverse magnetic (TM) solution of the equations, i.e., $B^z = 0$. Show that the x and y components of the total electric field may be expressed using the z component, $E^x = \partial_x \partial_z E^z / k_\parallel^2$, $E^y = \partial_y \partial_z E^z / k_\parallel^2$. Write down the electric field $\boldsymbol{E}$ explicitly using $E_0, \omega, k_\parallel$ and (k^x, k^y, k^z).

b) Consider the previous solution and assume now that the waves are evanescent along the z direction, $k^z = -\mathrm{i}\,|k^z|$. Show that in this case the electric field may be written as

$$\boldsymbol{E} = \mathrm{Re}\,\left(\boldsymbol{E}_r\left(k^x, k^y, k^z\right)\mathrm{e}^{\mathrm{i}\omega t}\right),$$

where all amplitudes are real $\boldsymbol{E}_r \in \mathbb{R}^3$, i.e., the components oscillate in the same phase.

c) Demonstrate that the amplitude of the electric field does not vanish anywhere in space for $k^z \neq 0$. This means that it can be characterized by its direction $\boldsymbol{E}_r / |\boldsymbol{E}_r| \in S^2$ everywhere.

<u>Exercise 10:</u> States on a discrete phase space

a) Consider a single classical Ising spin on the phase space $P = \{0, 1\}$. States are probability measures defined on P. Which topological space do the pure states form? Consider all mixed states, which are convex combinations of pure states. Which topological space do all states form together?

b) Generalize the above to a d-state Potts variable, $P = \{0, 1, \ldots, d-1\}$. Which topological space do the pure states form? What is the topological space of all states combined?

<u>Exercise 11:</u> Mixed states of a qubit

Consider a mixed state on the Hilbert space $\mathbb{C}^2$, $\varrho = \frac{1}{2}\left(\sigma_0 + \boldsymbol{d}\boldsymbol{\sigma}\right)$ with $\|\boldsymbol{d}\| < 1$. Show that the decomposition into the convex combination of pure states is not unique. In particular, for any pure state $\varrho_1 = 1/2\left(\sigma_0 + \boldsymbol{e}_1\boldsymbol{\sigma}\right)$ with $\|\boldsymbol{e}_1\| = 1$, find another pure state $\varrho_2 = 1/2\left(\sigma_0 + \boldsymbol{e}_2\boldsymbol{\sigma}\right)$ with $\|\boldsymbol{e}_2\| = 1$ such that $\varrho = p\varrho_1 + (1-p)\,\varrho_2$. Express $\boldsymbol{e}_2$ and p using $\boldsymbol{d}$ and $\boldsymbol{e}_1$.

<u>Exercise 12:</u> Stokes parameters and Poincaré sphere

Consider a transverse electric (TE) mode of an electromagnetic plane wave propagating along the z direction. That is,

$$\boldsymbol{E} = \mathrm{Re}\,\left(\boldsymbol{E}_0 \mathrm{e}^{\mathrm{i}(\omega t - kz)}\right) = \mathrm{Re}\,\boldsymbol{E}_0 \cos\left(\omega t - kz\right) - \mathrm{Im}\,\boldsymbol{E}_0 \sin\left(\omega t - kz\right),$$

where $\boldsymbol{E}_0 \in \mathbb{C}^2$ is a complex amplitude vector in the xy plane.

a) Introduce the Stokes parameters

$$\begin{aligned}S_0 =& \left|E_0^x\right|^2 + \left|E_0^y\right|^2,\\ S_1 =& \left(\left|E_0^x\right|^2 - \left|E_0^y\right|^2\right)/S_0,\\ S_2 =& 2\mathrm{Re}\,\left(\overline{E_0^x} E_0^y\right)/S_0,\\ S_3 =& 2\mathrm{Im}\,\left(\overline{E_0^x} E_0^y\right)/S_0.\end{aligned}$$

Show that the last three components define a vector of unit length, $(S_1, S_2, S_3) \in S^2$. The set of possible values of this vector is called the Poincaré sphere.

b) Find vectors $\boldsymbol{E}_0$ that result in certain points on the Poincaré sphere,

$$
\begin{aligned}
&S_1 = 1, S_2 = 0, S_3 = 0,\\
&S_1 = -1, S_2 = 0, S_3 = 0,\\
&S_1 = 0, S_2 = 1, S_3 = 0,\\
&S_1 = 0, S_2 = -1, S_3 = 0,\\
&S_1 = 0, S_2 = 0, S_3 = 1,\\
&S_1 = 0, S_2 = 0, S_3 = -1,\\
&S_1 = \frac{\sqrt{2}}{2}, S_2 = 0, S_3 = \frac{\sqrt{2}}{2}.
\end{aligned}
$$

c) For each of the selected points on the Poincaré sphere above, sketch the trajectory of the real-valued electric field $\boldsymbol{E}$ in the xy plane for a full period, $\omega t - kz \in [0, 2\pi]$. Analyze the shapes of teh observed trajectories. What is the meaning of the S_1 and S_2 parameters? How is the S_3 parameter connected to the other two? What does the sign of S_3 influence?